DO NOW — RECALL

Assembly language and addressing modes

Name: _____ Class: _____ Date: _____

1. Which primary memory is normally volatile?

2. What is the MAR used for?

3. What does an array index indicate?

4. For int[] a={5,8,3,7}, give a[2], then execute a[1]=a[1]+a[2]. State the new array and total.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

RAM.

2. Model answer

Holding the address of memory being accessed.

3. Model answer

The position of an element within an array.

4. Model answer

a[2]=3; a[1]=11; array [5,11,3,7], sum 26.

Vocabulary — know these terms

immediate — Operand value encoded as a literal.

direct — Memory address of data specified in instruction.

indirect — Specified memory holds a further effective address.

indexed — Base address modified by index or displacement.

effective address — Final memory location used for operand access.

NEW LEARNING

Read, annotate and explain

Literal versus location

Immediate addressing embeds a literal value; direct addressing points to the memory address holding a value.

Follow a pointer for indirect

M[21] contains 20, so an indirect read through address 21 accesses M[20]=7.

Calculate indexed address

Base 20 plus index X=2 gives effective address 22 and value 3. These operand formats are generic examples rather than LMC syntax.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

M[30]=4, M[31]=30, M[32]=9, X=2. Evaluate #6, [30], [[31]], [30+X].

Predict

Immediate is literal 6; direct gets 4 from memory address 30.

Explain

Indirect follows pointer 30 stored at 31 to value 4; indexed adds offset 2 to base 30 to read 9.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define immediate, direct, indirect and indexed addressing.

Question 2

Given M[10]=3, M[11]=10, M[12]=9 and X=2, find #4, [10], [[11]] and [10+X], explaining both address calculations.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

An array of 16-bit values begins at byte address 100. Find byte address of array[3] and explain why base+3 would be wrong for byte-addressed memory.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Indirect addressing returns the pointer value itself rather than following the pointer.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.